

Homework 05 Solutions

Question 1: Estimate the bias and variance of KNN

Solution a

► Show the solution code

	k	prediction	truth
1	1	0.7888302	0.5942177
2	3	0.6621674	0.5942177
3	5	0.6642853	0.5942177
4	7	0.6100117	0.5942177
5	9	0.6318671	0.5942177
6	11	0.4925114	0.5942177
7	13	0.5222989	0.5942177
8	15	0.5810203	0.5942177
9	17	0.5738171	0.5942177
10	19	0.5751479	0.5942177
11	21	0.5800517	0.5942177
12	23	0.5519177	0.5942177
13	25	0.5214556	0.5942177
14	27	0.4952957	0.5942177
15	29	0.4972667	0.5942177

...

Solution b

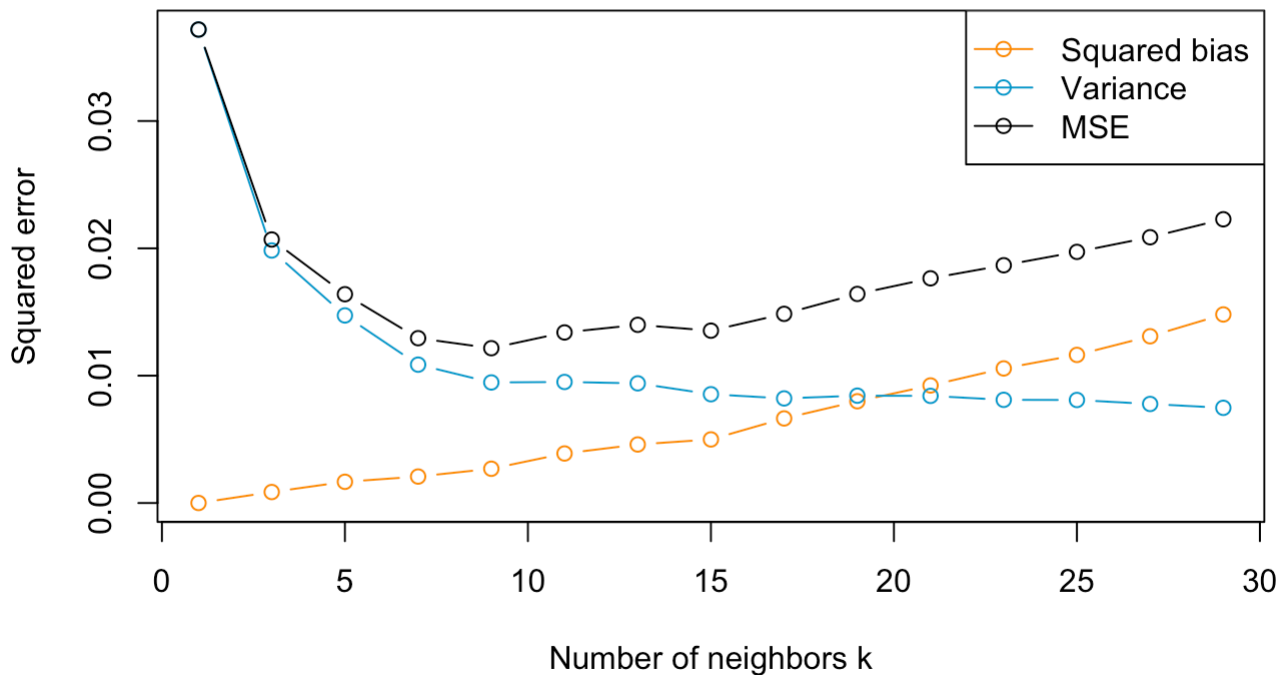
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	k	bias	variance	MSE
1	1	-0.001248923	0.037181204	0.03718276
2	3	-0.029361680	0.019831580	0.02069369
3	5	-0.040815728	0.014731018	0.01639694
4	7	-0.045589732	0.010865566	0.01294399
5	9	-0.051867114	0.009467440	0.01215764
6	11	-0.062360901	0.009506180	0.01339506
7	13	-0.067797856	0.009397702	0.01399425
8	15	-0.070662763	0.008542311	0.01353554
9	17	-0.081499864	0.008213727	0.01485595
10	19	-0.089385641	0.008429774	0.01641957
11	21	-0.096080776	0.008413451	0.01764497
12	23	-0.102803643	0.008103017	0.01867161
13	25	-0.107847900	0.008091147	0.01972232
14	27	-0.114428351	0.007777947	0.02087180
15	29	-0.121688194	0.007474576	0.02228259

► Show the solution code

[1] 6.765422e-17

► Show the solution code



Question 2: Compare KNN with lasso

Solution a

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Setting	KNN_MSE	Lasso_MSE	Lambda	Nonzero
Independent	0.5565	0.2602	0.0251	11
Correlated	0.1609	0.2259	0.0100	16

Solution b

**For independent covariates, lasso has the smaller test MSE than KNN. Because Lasso explicitly discards noise variable coefficients to zero. For correlated covariates, KNN has the smaller test MSE than Lasso. Because highly correlation helps distance metrics find meaningful neighbors.

Solution b

**KNN offers non-parametric flexibility (capturing non-linearities without explicit terms) but lacks feature selection. Lasso offers powerful feature selection but suffers when the true relation is highly non-linear.

Question 3: Observed dimension and latent dimension

Solution a

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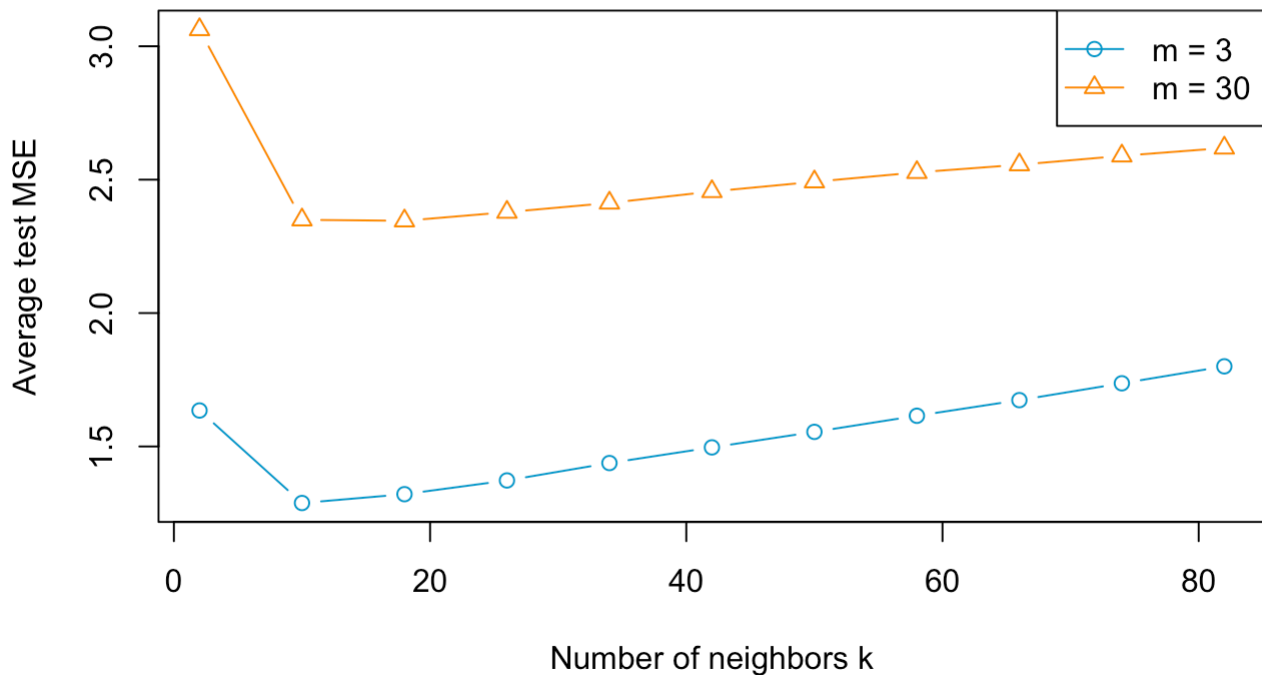
	k	m3	m30
1	2	1.449265	3.457456
2	10	1.253345	2.583372
3	18	1.292923	2.533071
4	26	1.379955	2.524038
5	34	1.448416	2.512187
6	42	1.505892	2.560783
7	50	1.590057	2.562911
8	58	1.660186	2.593970
9	66	1.739597	2.607162
10	74	1.816963	2.651181
11	82	1.871802	2.672724

Solution b

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	k	m3	m30
1	2	1.634822	3.062662
2	10	1.288062	2.349524
3	18	1.321139	2.345840
4	26	1.372628	2.378219
5	34	1.437850	2.412709
6	42	1.496497	2.455664
7	50	1.554715	2.492101
8	58	1.615062	2.527212
9	66	1.673644	2.556039
10	74	1.736733	2.589309
11	82	1.800274	2.618853

► Show the solution code



Solution c

**When $m = 3$, it is easy for KNN to find good matches, so its predictions are accurate. When $m = 30$, KNN is influenced by other irrelevant covariates, so it is hard to get the matching distance. The measurement noise represent unobserved variability and it scatters the data off the main subspace.

Question 4: Classify handwritten digits

Solution a

► Show the solution code

Solution b

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Euclidean test error: 0.122

	True									
Predicted	0	1	2	3	4	5	6	7	8	9
0	118	0	7	1	0	4	2	0	2	0
1	0	66	0	0	2	0	0	0	0	1
2	0	0	39	1	0	0	2	0	0	1
3	0	0	0	23	0	4	0	0	2	0
4	0	0	1	0	27	1	1	1	0	1
5	0	0	0	1	0	13	0	0	0	0
6	0	2	0	0	0	1	41	0	2	0
7	0	0	0	0	1	1	0	34	0	4
8	0	0	3	2	0	2	0	0	37	0

	9	1	0	0	1	4	0	0	1	1	41
Manhattan test error: 0.138											
True											
Predicted	0	1	2	3	4	5	6	7	8	9	
0	118	0	6	1	0	5	3	0	4	0	
1	0	67	2	0	2	0	1	0	0	1	
2	0	0	35	0	0	0	2	0	0	0	
3	0	0	1	24	0	4	0	0	2	0	
4	0	0	1	1	27	1	0	1	0	1	
5	0	0	0	1	0	11	0	0	0	0	
6	0	1	0	0	0	2	40	0	2	0	
7	0	0	2	1	1	0	0	34	0	5	
8	0	0	2	0	0	3	0	0	34	0	
9	1	0	1	1	4	0	0	1	2	41	

The test error is 0.122 for Euclidean distance and 0.138 for Manhattan distance.

Solution c

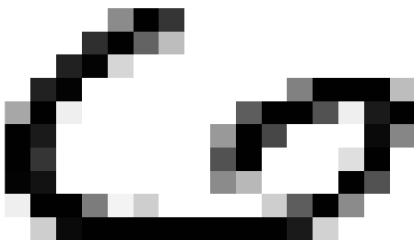
► Show the solution code

```
[1] "digit 6 taken"
```

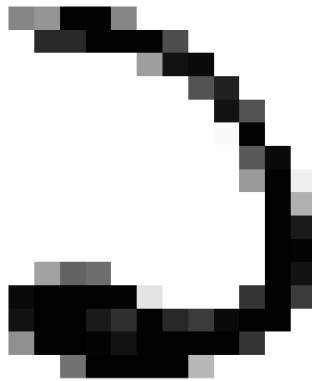
```
[1] "digit 2 taken"
```

```
[1] "digit 4 taken"
```

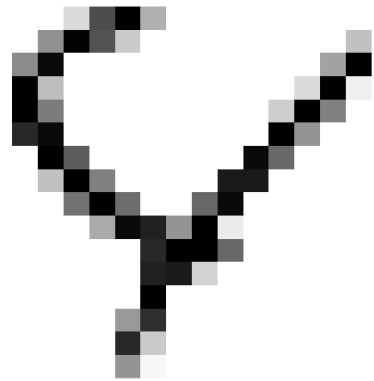
True: 6 Predicted: 0



True: 2 Predicted: 0



True: 4 Predicted: 7



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Different predictions: 22

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	Manhattan_correct	
Euclidean_correct	FALSE	TRUE
FALSE	58	3
TRUE	11	428